

Continued High Levels of Invasive *Streptococcus pyogenes* Infections in Belgian Children Under 5 Years

Exploring Different Contributing Factors

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Abstract

Background: Many countries saw an increase in invasive infections caused by *Streptococcus pyogenes* (iGAS) late 2022 to early 2023. Several hypotheses have been put forward to explain this increase, but thus far, reports have been limited to the immediate post-pandemic period, and the relative contribution of each of those factors remains unknown.

Methods: We describe the epidemiology of iGAS in Belgium in children below 5 years of age from January 2014 until December 2024 based on laboratory data from the National Reference Centre. Using a cointegration analysis after removing seasonality, we formally assess the influence of co-circulating viral diseases varicella, respiratory syncytial virus and influenza on iGAS case numbers.

Results: After very low levels during the pandemic years, a marked increase started in June 2022, reaching an all-time high in April 2023 before decreasing but still remaining at high levels in early 2024. Reduced population immunity alone cannot explain this pattern. *emm12*, *emm1* and *emm3* genotypes succeeded each other as the predominant genotype, indicating a complex interplay of transmission and virulence factors. Levels of varicella infections, but not respiratory syncytial virus or influenza, are closely linked to the increase in iGAS levels.

Conclusion: Levels of iGAS infections remained high up to 2 years after lifting nonpharmaceutical interventions, and this increase is undoubtedly multifactorial. Continued high-quality surveillance is required to follow up the situation.

Key Words: group A streptococcal infections, varicella, epidemiology, Belgium, vaccine-preventable diseases

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INTRODUCTION

Streptococcus pyogenes or group A streptococcus (GAS) causes an exceptionally wide variety of clinical presentations, from asymptomatic carriage over immune-related conditions to invasive disease.¹ The highest case fatality rates, up to 30% in high-income countries,² are for necrotizing fasciitis and streptococcal toxic shock syndrome (STSS), severe forms of invasive GAS disease (iGAS). iGAS is defined as the entry of GAS into a normally sterile site of the body, such as deep tissues or the bloodstream. Worldwide, iGAS infections cause around 163,000 annual deaths.³

Risk factors for invasive disease remain poorly understood and consist of both host and pathogen characteristics. On the host side, these include extremes of age, underlying chronic conditions, skin wounds and poverty.¹ On the pathogen side, there are a variety of virulence factors, such as the M protein, which is used to characterize GAS isolates by genotyping of its encoding *emm* gene. Today, more than 260 different variants (*emm* types) have been identified.⁴ Protective immunity might require repeated exposure to different M types to build broad protection. Alternatively, broad protection might be acquired by repeated exposure to conserved non-M-protein antigens.⁵ The absence of broad protective immunity likely explains the higher incidence of iGAS infections in children below 5 years of age, as compared with older children or adults.

In Europe, (i)GAS infections show a marked seasonality, with higher levels during late winter/early spring. This is also peak season for many other respiratory diseases, such as influenza, respiratory syncytial virus (RSV) or varicella. While factors such as humidity, temperature or time spent indoors might increase the risk of each of these infections independently, there is also evidence to suggest a direct interaction between influenza and iGAS,^{6–8} varicella and iGAS^{9–11}, or respiratory diseases in general and iGAS.^{12,13}

In late 2022, the United Kingdom and several European countries reported a markedly increased number of iGAS infections.^{14–16} Whereas the incidence of iGAS had been very low in 2020 and 2021, during the coronavirus disease (COVID) pandemic, 2022 levels were well above pre-pandemic averages. In many countries, the increase mostly affected young children.¹⁷ iGAS increases have been linked to the emergence of a more pathogenic strain, similar to the M1_{UK} in 2016 in the United Kingdom,¹⁸ but also to reduced immunity, as a result of limited exposure in the preceding years, or to increased viral coinfections.¹⁷ Thus far, reports are mostly available for the immediate post-pandemic period, without information on the further evolution in 2024. Therefore, we here describe the Belgian epidemiology of iGAS in young children (<5 years of age) until the end of 2024, and explore different factors that could contribute to the immediate post-pandemic increase.

METHODS

Study Period

We included data from January 1, 2014 to December 12, 2024. The pandemic period was defined as March 2020 until

February 2022, as this was the period where most nonpharmaceutical interventions (NPIs) were in place.¹⁹

iGAS Epidemiology

For *iGAS infections* in Belgium, we used data from the National Reference Centre (NRC). All Belgian labs are invited to send GAS isolates from a normally sterile site to the NRC, free of charge, for confirmation, antibiotic susceptibility testing and *emm* genotyping. Participation is voluntary but has been stable over time (± 75 labs/year) and covers patients from all Belgian provinces.²⁰ Patient's date of birth, clinical presentation and vital status are transferred together with the isolate through a standardized request form. Only 1 clinical presentation (the most severe form) can be reported per case. Isolates that did not contain *Streptococcus pyogenes*, were noninvasive, or repeated isolates from the same patient were excluded. Until 2018, susceptibility to erythromycin was determined by disk agar diffusion for all submitted strains. From 2019 onward, a yearly random selection of 50 strains (with wide geographical distribution) was made.

We calculated case counts by month, pandemic period, age group and *emm*-type. Proportions were compared using Fisher exact test.

Temporal Association with other Diseases

To investigate the hypothesized association between iGAS in children <5 years of age on the one hand and varicella and respiratory pathogens on the other hand, we used monthly time-series data.

For *varicella*, Google Trends (trends.google.com) data were used. These reflect the proportion of searches for a given keyword in a specific region and time, compared with all searches for all topics during that period and from that region. The source has been previously validated for varicella surveillance in international peer-reviewed publications²¹ as well as specifically for Belgium by comparison with data from a network of sentinel general practitioners.²² Although these data do not give information on the age of the patients, both incidence data from the general practitioners sentinel network as well as seroprevalence data show that >80% of Belgian varicella cases occur in children <5 years of age.²³

For *respiratory infections*, we used data on influenza and RSV from a network of sentinel labs.²⁴ Participating labs send monthly data on positive tests to the public health institute. Twenty-seven labs, geographically spread throughout Belgium,

contributed data on RSV and influenza in children <5 years of age, for each year of the study period. Although these are not the only pathogens that can cause respiratory infections in children, previous research in Belgium has shown that 78% of children <5 years presenting to the hospital with a severe acute respiratory infection tested positive for RSV.²⁵

The presence of a seasonal trend would invalidate the methods described later. We therefore first removed it from the data: we applied a seasonal-trend decomposition using LOESS smoothing with a fixed seasonal component to the pre-COVID subseries of all 4 diseases, to then extend and subtract these seasonal patterns over the entire study period. We estimated seasonal trends solely on pre-COVID data because the dynamics of the diseases under investigation changed during the COVID pandemic. We then tested for cointegration using the Engle–Granger 2-step approach.²⁶

In a first step, we assessed stationarity of each time series using augmented Dickey–Fuller and Kwiatkowski–Phillips–Schmidt–Shin tests and visual inspection. For stationary series, statistical properties such as mean, variance and autocorrelation structure remain constant over time. Nonstationary series, by contrast, exhibit trends, random walks (unit roots) or structural breaks that prevent the series from returning to a constant mean value. Note that there cannot exist a linear relationship between a nonstationary and a stationary series. In a second step, if the iGAS series was nonstationary, we tested for cointegration of the nonstationary variables which would indicate the presence of a stable, long-run equilibrium rather than a spurious correlation. If the iGAS series was stationary, we regress it on the other stationary disease series. Finally, we build an error correction model for the cointegrated time series, to describe how quickly deviations from the long-term equilibrium are restored.

RESULTS

We included a total of 679 iGAS cases in children under 5 years of age. Figure 1 shows monthly case numbers of iGAS by age group. In the pre-pandemic years, monthly case numbers peaked in March–April with an annual average of 46 cases per year for children <5 years (range 26–60) and 306 for all ages combined (range 227–376). After very low case numbers in pandemic years 2020 and 2021, there was an atypical sudden rise in June 2022. An initial decrease was then followed by a massive increase from October 2022 onwards, peaking at the end of 2022 to early 2023. Annual case numbers for children were higher than the pre-pandemic average of 46 per year for all 3 post-pandemic years,



FIGURE 1. Monthly case numbers of invasive infections caused by *Streptococcus pyogenes* and proportion in children <5 years, Belgium, January 2014 to December 2024.

at 127 (2022), 164 (2023) and 67 (2024). Children <5 years made up, on average, 15% of all cases from 2014 until the end of 2019. However, their proportion rose to 22% in 2022 ($P < 0.001$ χ^2 test), to then again normalized in 2023 (17%) and finally decreased to only 11% in 2024 ($P = 0.01$ for comparison with pre-pandemic proportion).

Figure 2 shows the different clinical presentations of iGAS cases in children <5 years of age and their evolution over time. In general, the distribution was similar between the pre- and post-pandemic periods. Most cases presented as sepsis (364/679, 54%), both pre- (56%) and post-pandemic (50%). There was a notable peak of cases presenting with pneumonia (9 of 27 reported cases that month) in December 2022. However, proportions of cases presenting with pneumonia for the whole of 2022 (13/127, 10.2%), 2023 (14/163, 8.6%) or the entire post-pandemic period (29/349, 8.3%) were not different from the pre-pandemic period (25/285, 8.8%).

Case fatality in children was similar in the pre-pandemic period (15 deaths of 285 cases, 5.3%) compared with the post-pandemic period (9/349 = 2.6%, $P = 0.1$).

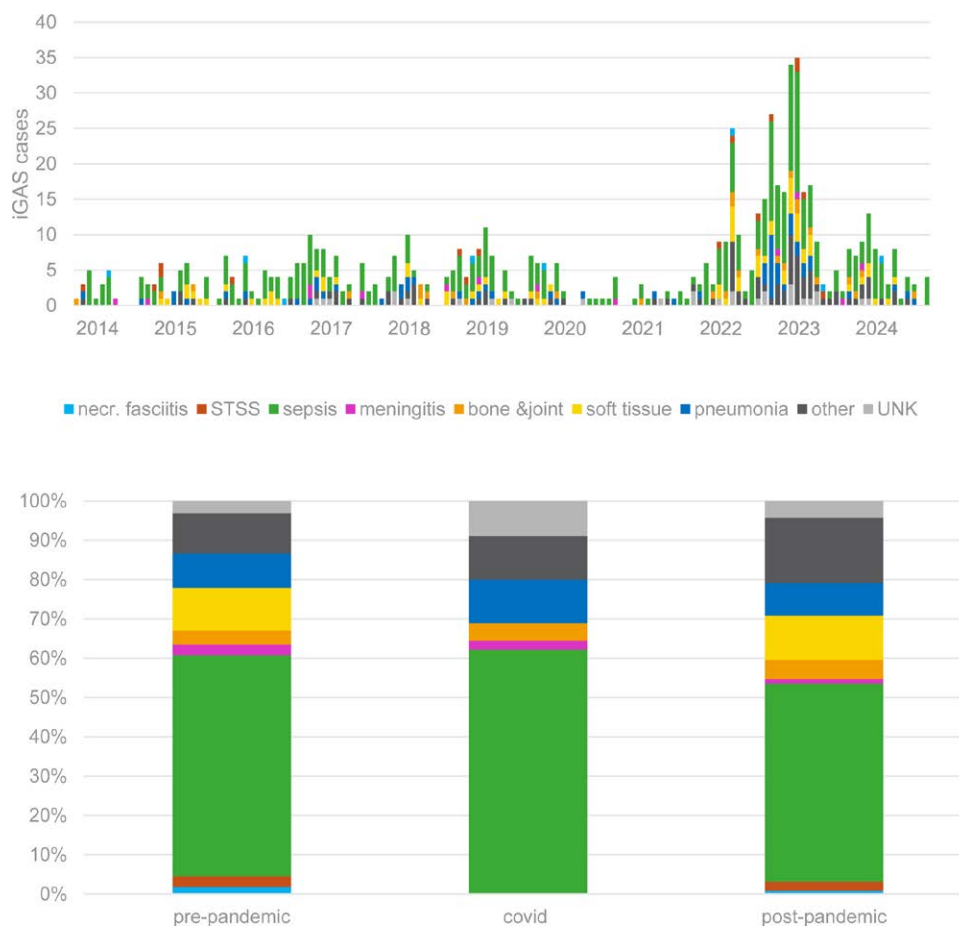
The distribution of *emm* types among iGAS cases is given in Figure 3. Overall, *emm1* was the most frequent genotype (228/679,

34%) both pre- (34%) and post-pandemic (36%). However, only very little *emm1* circulated during the pandemic, when most cases (14/44, 32%) were caused by *emm87*. Of note is that the post-pandemic period can be subdivided into different waves. The peak in June 2022 consisted of 52% *emm12* cases. Over the following months, *emm1* became the most frequent genotype again, but with an increasing proportion of *emm4* (13/35 cases in April 2023). Finally, in 2024, most cases were caused by *emm3* (8/14 cases for the peak in March 2024).

Erythromycin resistance remained stable around 4%–9% throughout the study period.

Figure 4 shows the evolution over time of iGAS case numbers in children <5 years and viral infections. The time series included 47,877 positive tests in children <5 years for RSV and 31,417 for influenza. Notably, both iGAS and varicella infections peaked in June and showed increased levels at the end of 2022 and throughout 2023. In contrast, iGAS infections remained low during the seasonal RSV peak in Spring 2021. For influenza, the post-pandemic annual epidemic peaks were not aligned with the iGAS peaks.

Stationarity testing yielded varying results depending on modeling assumptions for iGAS, varicella, and RSV. After visual



STSS = streptococcal toxic shock syndrome ; UNK = unknown

FIGURE 2. Clinical presentations of children <5 years with iGAS, Belgium, January 2014 to December 2024, in absolute numbers per month (upper panel) or proportions per period (lower panel). STSS indicates streptococcal toxic shock syndrome; UNK, unknown.

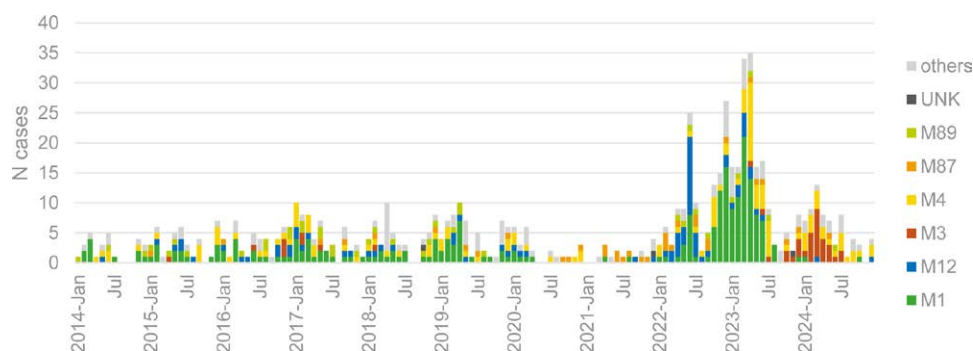


FIGURE 3. Distribution of *emm* types among iGAS cases in children <5 years of age, Belgium, January 2014 to December 2024.

inspection of the series (plots in Supplemental Digital Content 1, <https://links.lww.com/INF/G539>), we concluded that iGAS and varicella demonstrated nonstationary behavior, and RSV and influenza demonstrated stationary behavior, with some uncertainty about RSV. As the deseasonalized iGAS time series was nonstationary, this precludes a valid linear regression with the stationary series RSV and influenza, so we only tested cointegration between iGAS and varicella. We did indeed find a significant cointegrating relationship between these two infections. The error correction model demonstrated a strong error correction mechanism with a coefficient of -0.48 (95% confidence interval $[-0.63; -0.33]$, $P < 0.001$). This indicates that approximately 48% of deviations from the long-run equilibrium are corrected each period, suggesting a robust tendency for these disease patterns to maintain their relationship. Because stationarity was doubtful for RSV, we repeated the cointegration analysis but additionally included RSV. However, inclusion of RSV did not alter the long-term equilibrium between iGAS and varicella, and its coefficient in the linear equilibrium was not statistically significant.

DISCUSSION

We describe the epidemiology of invasive *S. pyogenes* infections in children over a period of 11 years, spanning multiple years before and after the COVID-19 pandemic. Typical seasonal patterns were severely disrupted during and immediately after the pandemic, with atypical peaks for children <5 years in June and December 2022 and extremely high case numbers in March 2023. Since then, normal seasonality seems to have been restored, although absolute case numbers remained high even in 2024. Overall, the increase was not associated with a change in clinical presentation or severity. This is in contrast with reports from several other countries that reported higher proportions of pneumonia and empyema.^{15,16,27,28} The difference might be explained by different time periods and an interplay of several dynamics. During the peak in June 2022, which coincided with high varicella activity, many skin and soft tissue infections were observed. In contrast, in December 2022, when RSV and influenza activity were high, more complicated pneumonia was reported. Moreover, most cases were reported as sepsis, without information on the initial presentation. Some of these cases might initially have presented as pneumonia and subsequently deteriorated. While only very few cases of STSS have been reported, a previous Belgian study found a significant underdiagnosis of STSS in medical records.²⁹ The reported case fatality of 3%–5% is similar to the 5.1% found in Spain³⁰ and highlights the severity of these infections.

While the observed increase is clear, the underlying reasons remain much more elusive. An increased pool of susceptible

children, because of near absence of iGAS infections during the pandemic, likely played an important role. Indeed, a decline in GAS antibodies during the pandemic has been demonstrated in a study in New Zealand.³¹ In our study, the increase was initially more pronounced in the youngest children, as shown by the higher proportion of cases <5 years in 2022. As some form of protective immunity is typically built up in the first years of life by repeated exposure to different types of GAS, reduced GAS circulation would have particularly impacted the youngest age groups. Interestingly, despite very high levels of iGAS end of 2022/early 2023, the increase continued end of 2023 and in 2024, and the relative contribution of the younger children was then actually lower than pre-pandemic. Additionally, while the pandemic years were characterized by very few cases of iGAS, we cannot simply conclude that exposure to GAS was equally reduced without data on asymptomatic carriage. For *Streptococcus pneumoniae*, another *Streptococcus* whose clinical presentation can range from asymptomatic carriage to life-threatening invasive disease, it has been shown that whilst invasive disease was nearly absent during the pandemic, levels of carriage were not much impacted.³² Finally, as de Gier et al. argue, generally increased susceptibility should have equally caused increases in non-invasive GAS, which the authors did not observe in the Netherlands.³³ We must thus investigate other potential factors for the increase.

Several publications, including one from Belgium, have pointed to an increase in the M1_{UK} lineage, which is more virulent due to enhanced SpeA expression.^{34–36} However, we found that the post-pandemic period of increased circulation could be divided into 3 waves that were predominated by first *emm12*, then *emm1* and, finally, *emm3*. An increase of M1_{UK} alone is hence unlikely to explain the increased circulation. Similar to our findings, high proportions of *emm12* have also been reported in Norway and Spain.^{16,27} In contrast, in Denmark, increased case numbers were accompanied by the appearance of a rapidly expanding novel sub-lineage, named M1_{DK}. As this lineage has no known pathogenic mutation, the association is more likely due to clonal expansion than to a fitness advantage or higher intrinsic virulence.³⁷ Indeed, increased numbers of invasive disease caused by a certain genotype can be the result of either increased virulence, ‘founder effect’, reduced immunity, a transmission advantage or a combination of all of these factors. With regards to transmission, it has been described that preferred tissue sites differ by *emm* cluster, with genotypes that belong to the A-C cluster being ‘throat specialists’.³⁸ These throat specialists, including *emm1*, *emm12* and *emm3* might rely particularly on respiratory transmission³⁴ and hence be both more impacted by NPIs like mask-wearing as well as more able to rapidly spread when NPIs are lifted. Moreover, as Rümke et al.³⁵ postulate, SpeA-specific immunity might have been reduced post-pandemic

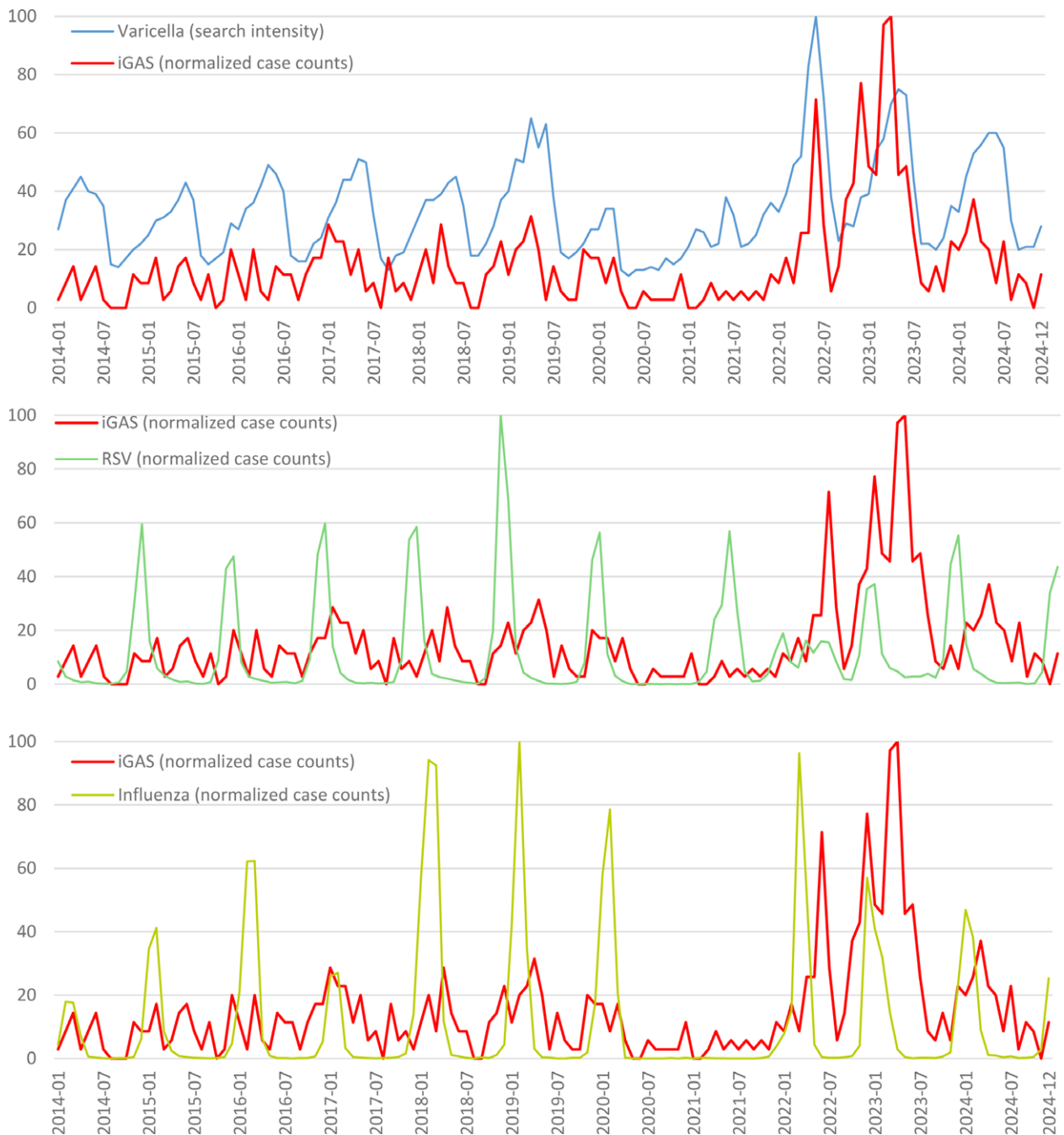


FIGURE 4. Evolution of monthly number of iGAS cases and circulation of other viral infections in children <5 years of age, Belgium, January 2014 to December 12/2024.

because of lower levels of non-*emm1 speA*-carrying types, notably *emm3* and *emm6*, in the preceding years.

Finally, increased co-circulation of viral infections might have contributed to increased case numbers. In our study, we did not find a convincing association between increased GAS and either influenza or RSV circulation. This is in contrast with what has been previously described. A recent modeling study from the Netherlands found varicella, influenza A, RSV and hMPV all contributing to GAS pneumonia and sepsis in children 0–5 years of age.³³ However, that study used a different methodology. We used

a robust cointegration analysis and subtracted the seasonal trend to avoid a spurious relationship based on shared seasonality rather than a true causal association. We did indeed find a strong association between iGAS and varicella, which was also by far the main contributor in the Dutch modeling study. The large error correction coefficient is biologically plausible, as varicella lesions can act as a port of entry for the group A streptococci. Varicella vaccination is currently not included in the routine national vaccination program in Belgium. Although relatively rare, the severity of iGAS cases might constitute an argument in favor of introducing such

a vaccine. Indeed, countries like Canada and Israel have reported a decrease in iGAS incidence after the introduction of varicella vaccination.^{39,40}

Our study has some limitations. As it is based on lab surveillance only, with clinical information collected at the time of sending of the sample, there might be an underestimation of culture-negative cases and case fatality. As Belgium is a relatively small country and we only looked at children under 5 years of age, case numbers were relatively low and needed to be grouped into monthly counts before modeling. Although the number of labs that sent positive samples to the NRC remained relatively stable over the years, we cannot rule out that part of the observed increase is due to heightened awareness resulting in increased case finding. However, our findings are concurrent with reports from several other countries. The prolonged observation period, spanning multiple years before and after the pandemic, constitutes a decided strength of the study. We moreover offer a comprehensive exploration of different factors that could underlie this increase.

CONCLUSION

We showed that, like in several other countries, iGAS cases in Belgian children <5 years of age surged after the pandemic, probably owing at least in part to a reduced population immunity. Interestingly, however, levels in 2024 were still above what would have been expected based on circulation in 2014–2019, and the *emm3* genotype replaced *emm1* early in 2024. Cointegration analysis strongly suggests that circulation of varicella, but not of RSV or influenza, significantly influenced levels of iGAS. Introducing varicella vaccination could hence aid to reduce iGAS infections. Surveillance, including genotyping, should be continued to follow up on the situation.

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